GIS 6103 - GIS Programming and Customization

1. OVERVIEW:
This hands-on graduate course provides a basic understanding of how to enhance and automate functional capabilities of a Geographic Information System (GIS) through Python scripting. Exercises and examples will primarily use geospatial analysis tools from within the ArcGIS Pro environment and a variety of Python libraries to extend the range of operations for GIS analysis.

- Fall semester, 3 credits
- 100% online, synchronous and asynchronous component
- http://elearning.ufl.edu/

Course prerequisites: While there are no formal course prerequisites, this class assumes that students have a basic knowledge of ArcGIS Pro since this is not an introductory GIS course. Previous programming experience with any language (Visual Basic, R, C, Java, or JavaScript), but preferably in Python will be very helpful.

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- Please use the Canvas message/Inbox feature for fastest response

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Lectures:
Fridays, 11:45am-2:30pm (per. 5-7) via Zoom; links to recordings are provided on the course website
First day of class: Fri, Aug 23; last day of class: Fri, Nov 22

Office hours:
Thursdays from 5-7pm in Zoom

Required reading materials:

- No course book is required

Recommended reading materials:

- Zandbergen, P. A. (2020). Python Scripting for ArcGIS Pro. Redlands, CA: ESRI Press.

Further optional reading materials:

- Zandbergen, P. A. (2020). Advanced Python Scripting for ArcGIS Pro. Redlands, CA: ESRI Press.
- Short instructional videos on the UF-FLREC Geomatics YouTube Channel

Additional materials:
- ArcGIS Pro Python reference. Available at
https://pro.arcgis.com/en/pro-app/arcpy/main/arcgis-pro-arcpy-reference.htm
- ArcGIS Pro tool reference. Available at
https://pro.arcgis.com/en/pro-app/tool-reference/main/arcgis-pro-tool-reference.htm

Software requirements:

- Latest version of ArcGIS Pro. Download and installation instructions are provided on under the Week 1 course module.
- Latest version of PyCharm Community edition; download at
https://www.jetbrains.com/pycharm/download/#section=windows
Note: Scroll down to bottom of page to find the Community Edition download option
- See instructions below on how to access ArcGIS Pro and PyCharm on UF Apps as an alternative.
- QGIS (free GIS software): https://qgis.org/download/

2. LEARNING OUTCOMES

The course objective is to provide students with the following competencies at the completion of the course:

1. Apply fundamental concepts of programming, such as loops and logical expressions
2. Manage and modify GIS datasets through ArcPy functions
3. Automate vector and raster data analysis through geoprocessing tools
4. Create Python based geoprocessing tools
5. Perform geodata processing and visualization within notebooks
6. Use Python libraries and GIS plugins
7. Use AI tools for assistance with programming

The course Website (see under Modules/Course Overview) contains an alignment table which illustrates how course activities (e.g., assignments, discussions) are linked to course competencies.

3. COURSE LOGISTICS

- For each assignment, discussion, and survey, a due date and time is given, which is the beginning of the next class (11:45am Fridays).
- Assignments are graded based on timeliness, completeness and correctness with written feedback by the instructor. The topic related discussion item is based on completeness and timeliness; the grading of the survey is based on completion and timeliness, and points for assignment-related online discussion posts are based on the number of contributions .
- There is a 1-week turnaround for assignment, discussion, and survey grading.
- This course is a distance education course taught as live lectures using the virtual classroom software Zoom. Lecture materials can be downloaded from weekly modules on the Canvas website.
- This course does not offer extra credit tasks

The Canvas system should be used as the platform for written communication between students and the instructor. Questions and suggestions to the whole class can also be posted under the Discussions tab. Any short-term changes

concerning lectures or other course components will be announced through Canvas. Feel free to call the instructors with any questions.

Technology Requirements:

- A computer or mobile device with high-speed internet connection and a headset and/or microphone and speakers to view lectures or join live sessions
- ArcGIS Pro runs only on Microsoft operating systems. Students using a Mac computer or other operating systems may consider using UF Apps (https://info.apps.ufl.edu/) instead which has ArcGIS Pro and PyCharm installed. Though this is an alternative, the instructor cannot provide any technical support when choosing that option.
- For Zoom: A supported web browser on a supported operating system (Windows, Mac OS, Linux); and minimum bandwidth. More details can be found here.
- Canvas supports various browsers software, including Chrome, Firefox, Edge, and Safari, and JavaScript must be enabled. More information can be found here.

Using Zoom:

Live lectures and office hour (Q&A) meetings are conducted on Zoom. Sessions can be joined by clicking a link posted by the instructor on Canvas.

Synchronous online sessions will be recorded. By sharing your video, screen, or audio during any synchronous online class sessions, you are consenting to being recorded for the benefit of students who cannot attend live as well as for class review during the current semester. If you have special circumstances or concerns about privacy, it is your responsibility to discuss it with your instructor.

Grades:

Item	***Percentage***
Timeliness and completeness of assignments (12 assignments @ 7% each)	84%
Group discussion	5%
Survey	3%
Contribution to assignment-related online discussion posts (*)	8%
Total	**100%**

(*) provide at least one discussion post and one reply for five different home assignments for full credit

Grading scale:

Grade	***Percentage***	***Grade***	***Percentage***
A	92.0-100.0	C+	78.0-79.9
A-	90.0-91.9	C	72.0-77.9
B+	88.0-89.9	C-	70.0-71.9
B	82.0-87.9	D	60.0-69.9
B-	80.0-81.9	E	0-59.9

For information on current UF policies for assigning grade points, see
https://catalog.ufl.edu/ugrad/current/regulations/info/grades.aspx

4. COURSE CONTENT

Week	*Topic*	*Assessment*	*Readings*
Week 1, Aug 23	Introduction to Python resources and editors	H1	Intro book ch. 1-3
Week 2, Aug 30	Syntax, variables, control structures, objects	H2	Intro book ch. 4.5, 4.23
Week 3, Sep 6	Data types, file handling	H3	Intro book ch. 4.3-4.18, 8.7
Week 4, Sep 13	Intro to Arcpy and geoprocessing in Python	H4	Intro book ch. 5
Week 5, Sep 20	Spatial data management	H5	Intro book ch. 6
Week 6, Sep 27	Cursors – Feature selection and table manipulation	H6	Intro book ch. 8
Week 7, Oct 4	Feature geometries	H7	Intro book ch. 9
Week 8, Oct 11	Raster processing	H8	Adv. book ch. 2
Oct 18	*Homecoming (no class)*		
Week 9, Oct 25	Script tools	H9	Adv. book ch. 3
Week 10, Nov 1	Notebooks	H10	Adv. book ch. 7, 9
Week 11, Nov 8	Advanced python libraries for spatial analysis	H11	Adv. book ch. 6
Nov 15	Q&A session (no lecture)	D1	
Week 12, Nov 22	Open-source geospatial scripting	H12, S1	Online resources
Nov 29	*Thanksgiving (no class)*		

H: Home assignment, D: Discussion (separate from weekly assignment-related discussions), S: Survey

Grades and grade points:

For information on current UF policies for assigning grade points, see
https://catalog.ufl.edu/ugrad/current/regulations/info/grades.aspx

5. POLICIES AND REQUIREMENTS

This syllabus represents current plans and objectives for this course. As the semester progresses, changes may need to be made to accommodate timing, logistics, or to enhance learning. Such changes, communicated clearly, are not unusual and should be expected.

Late submissions and make-up requests:

It is the responsibility of the student to access on-line lectures and readings and to maintain satisfactory progress in the course.

- A 10% penalty per day will be applied to late assignments. A late submission on the due date results also in a 10% deduction.
- Assignments will not be accepted if handed in more than seven days after the due date.
- Online discussions and surveys cannot be completed past the deadline.
- Exceptions to the late policy are only allowed per university policy.

Computer or other hardware failures, except failure of the UF canvas system, will not excuse students for missing assignments. Any late submissions due to technical issues MUST be accompanied by the ticket number received from the Helpdesk when the problem was reported to them. The ticket number will document the time and date of the problem. You MUST e-mail your instructor within 24 hours of the technical difficulty if you wish to request consideration.

For computer, software compatibility, or access problems call the HELP DESK phone number—352-392-HELP = 352-392-4357 (option 2).

Semester Evaluation Process:
Student assessment of instruction is an important part of efforts to improve teaching and learning.

At approximately the mid-point of the semester, the SFFGS will request anonymous feedback on student satisfaction on various aspects of this course. These surveys will be sent out through Canvas and are not required, but encouraged. This is not the UF Faculty Evaluation!

At the end of the semester, students are expected to provide professional and respectful feedback on the quality of instruction in this course by completing course evaluations online via GatorEvals. Guidance on how to give feedback in a professional and respectful manner is available at https://gatorevals.aa.ufl.edu/students/. Students will be notified when the evaluation period opens, and can complete evaluations through the email they receive from GatorEvals, in their Canvas course menu under GatorEvals, or via https://ufl.bluera.com/ufl/. Summaries of course evaluation results are available to students at https://gatorevals.aa.ufl.edu/public-results/

Netiquette: Communication Courtesy Semester Evaluation Process:
All members of the class are expected to follow rules of common courtesy in all email messages, threaded discussions and chats, as laid out in the UF Netiquette Guide for Online Courses. Failure to do so may result in loss of participation points and/or referral to the Dean of Students' Office.

Diversity Statement:
This course encourages open and respectful communication among students, fostering a culture of inclusivity, understanding, and empathy. It values the unique backgrounds and experiences that each student brings to the classroom, including diverse racial and ethnic backgrounds, genders, sexual orientations, ages, abilities, religions, and socioeconomic statuses.

Academic Honesty Policy:
As a student at the University of Florida, you have committed yourself to uphold the Honor Code, which includes the following pledge: "*We, the members of the University of Florida community, pledge to hold ourselves and our peers to the highest standards of honesty and integrity*."

You are expected to exhibit behavior consistent with this commitment to the UF academic community, and on all work submitted for credit at the University of Florida, the following pledge is either required or implied: "*On my honor, I have neither given nor received unauthorized aid in doing this assignment*."

It is assumed that you will complete all work independently in each course unless the instructor provides explicit permission for you to collaborate on course tasks (e.g. assignments). Furthermore, as part of your obligation to uphold the Honor Code, you should report any condition that facilitates academic misconduct or appropriate personnel. It is your individual responsibility to know and comply with all university policies and procedures regarding academic integrity and the Student Honor Code. Violations of the Honor Code at the University of Florida will not be tolerated.

Violations will be reported to the Dean of Students Office for consideration of disciplinary action. For more information regarding the Student Honor Code, please see here. View this video for more information on how to avoid plagiarism.

University Policy on Accommodating Students with Disabilities:
Students requesting accommodation for disabilities need to request an accommodation letter from https://disability.ufl.edu/. The student then needs to provide this documentation to the instructor when requesting accommodation. Students need to submit this documentation prior to submitting assignments. Accommodations are not retroactive; therefore, students should contact the office as soon as possible in the term for which they are seeking accommodations.

Software use:
All faculty, staff and students of the university are required and expected to obey the laws and legal agreements governing software use. Failure to do so can lead to monetary damages and/or criminal penalties for the individual violator. Because such violations are also against university policies and rules, disciplinary action will be taken as appropriate.

Lecture recordings:
All live lectures and Q&A sessions will be recorded and made available via Canvas. Policies regarding student in-class recordings are detailed here http://aa.ufl.edu/policies/in-class-recording/.

6. CAMPUS RESOURCES

Academic Resources:

- For issues with technical difficulties for e-learning in Canvas, please post your question to the Technical Help Discussion in your course, or contact the UF Help Desk at:
  Learning-support@ufl.edu | (352) 392-HELP - select option 2 | https://helpdesk.ufl.edu/
- SFFGS Academic Hub https://ufl.instructure.com/courses/303721
- Career Connections Center: Reitz Union Suite 1300, 352-392-1601. Career assistance and counseling services.
- Library Support: Various ways to receive assistance with respect to using the libraries or finding resources.
- Writing Studio: 2215 Turlington Hall, 352-846-1138. Help brainstorming, formatting, and writing papers.
- Students Complaints: View resources here.

Health and Wellness:

- *U Matter, We Care*: If you or someone you know is in distress, please contact umatter@ufl.edu, 352-392-1575, or visit U Matter, We Care website to refer or report a concern and a team member will reach out to the student in distress.

- *Counseling and Wellness Center*: Visit the Counseling and Wellness Center website or call 352-392-1575 for information on crisis services as well as non-crisis services.
- *Student Health Care Center*: Call 352-392-1161 for 24/7 information to help you find the care you need, or visit the Student Health Care Center website.
- *University Police Department*: Visit UF Police Department website or call 352-392-1111 (or 9-1-1 for emergencies).
- Career Resource Center http://www.crc.ufl.edu/
- GatorWell Health Promotion Services https://gatorwell.ufsa.ufl.edu/